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Paper Title: A Data-Driven Approach to Forecasting Crop Transplantation Periods in Bangladesh Using Machine Learning Model

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OUTLINE OF PRESENTATION

- Introduction
- Objectives
- Methodology
- Data Analysis
- Result and Discussion
- Model Verification
- Conclusion
- Reference

INTRODUCTION

- Agricultural production in Bangladesh is extremely sensitive to climate variability. With increasing climatic uncertainties, evidence-based planting decisions are essential to sustain crop yields and ensure food security. [1]

- Climate change presents one of the greatest challenges to identifying optimal planting seasons, especially as historical calendars may no longer align with shifting environmental conditions. [2]

- Machine learning offers a promising approach to predict Optimal Planting Seasons. [3]

OBJECTIVES

The main goal of this study is to develop an intelligent system that can predict the optimal planting season for Bangladeshi agriculture using machine learning techniques. This will help farmers move beyond traditional methods and rely on more accurate and reliable approaches for cultivation.

METHODOLOGY

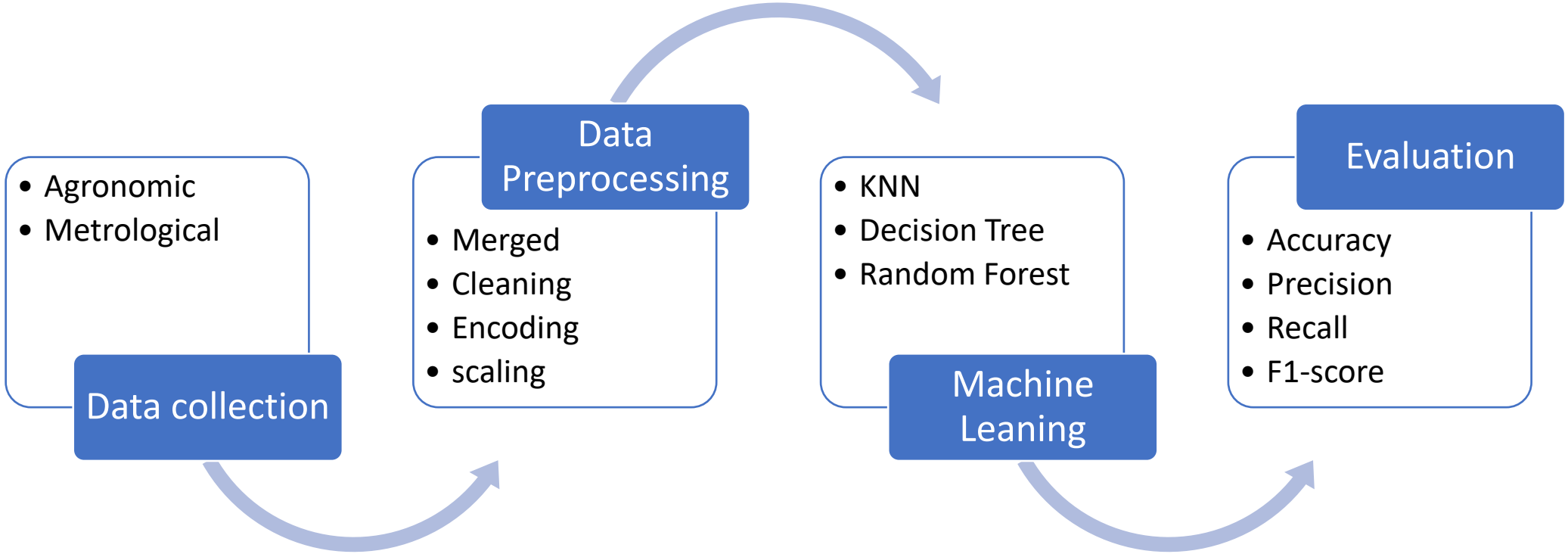


Fig. 1. Methodology of this Study

DATASET

Dataset	Variables	Dataset	Variables
Agronomic [4]	Crops Name Crops Type Max Temperature Min Temperature Relative Humidity Transplantation Month	Metrological [5]	Year Rainfall Sunshine Humidity Temperature

Table 1. Dataset Description.

DATA ANALYSIS

Crop Type	No of Crops	Max Temp.	Min Temp.	Max Humidity	Min Humidity
Cereals	8	40	12	95	50
Fiber	4	40	18	80	43
Fruits	27	47	5	95	40
Nicotine	1	20	15	85	50
Oil Seeds	10	35	10	95	30
Pulses	5	35	10	95	20
Staple	1	32	27	75	65
Tubers	1	26.5	15	90	85
Vegetables	46	45	5	100	20

Table 2. Different crop types weather variation.

DATA ANALYSIS

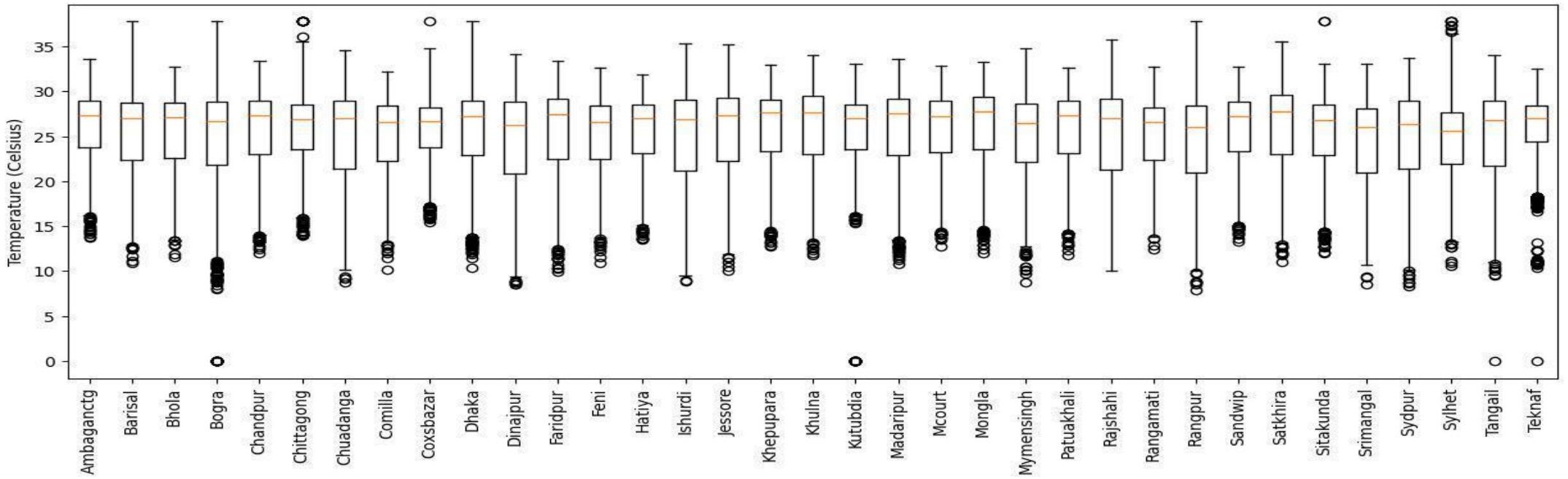


Fig. 2. Location-based Temperature in Bangladesh with IQR.

DATA ANALYSIS

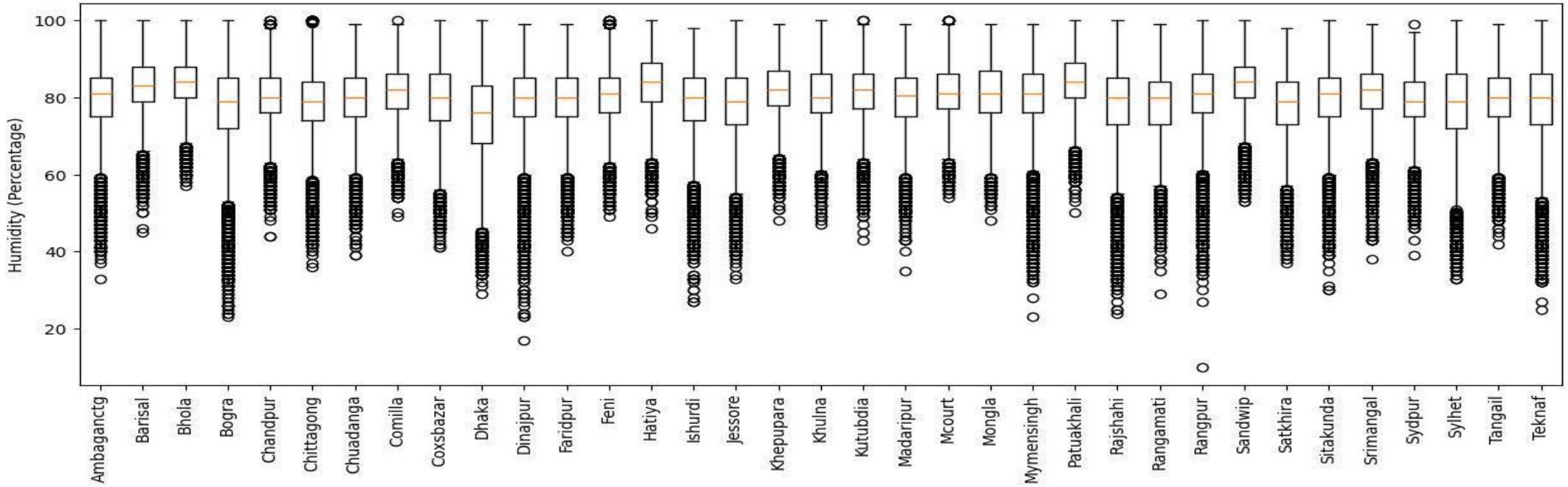


Fig. 3. Location-based Humidity in Bangladesh with IQR.

HYPER PARAMETER

Proposed Model	Location	Features
KNN	N_neighbors Metric	7 Minkowshi
Decision Tree (DT)	Max_depth	6
Random Forest (RF)	N_estimators	50

Table 3. Finalized Hyperparameters Used in Classification.

RESULT

Duration	Location	Features	KNN	Decision Tree	Random Forest
30 Years	YES	Metrological and Agronomic	97.68%	81.39%	98.95%
30 Years	YES	Metrological	90.13%	77.43%	98.86%
61 Years	NO	Metrological and Agronomic	97.43%	93.36%	98.56%
61 Years	NO	Metrological	93.78%	91.08%	98%

Table 4. Machine Learning Model 10-fold cross- validation mean accuracy based on different features.

RESULT

Month	KNN			RF			Support
	Precision	Recall	F1-score	Precision	Recall	F1-score	
January	1.00	1.00	1.00	0.99	0.98	0.98	558
February	1.00	1.00	1.00	1.00	1.00	1.00	1116
March	0.98	0.99	0.98	0.99	1.00	0.99	2046
April	0.99	0.98	0.98	1.00	1.00	1.00	3348
May	0.91	0.84	0.87	1.00	1.00	1.00	186
June	1.00	0.99	0.99	0.99	0.99	0.99	558
July	0.99	1.00	0.99	1.00	1.00	1.00	744
August	1.00	1.00	1.00	1.00	0.99	0.99	930
September	0.99	1.00	0.99	1.00	1.00	1.00	3720
October	0.99	0.99	0.99	0.99	0.98	0.99	3720
November	1.00	0.99	0.99	1.00	1.00	1.00	1488
December	1.00	1.00	1.00	1.00	1.00	1.00	744

Table 5. Classification report for KNN and RF.

RESULT

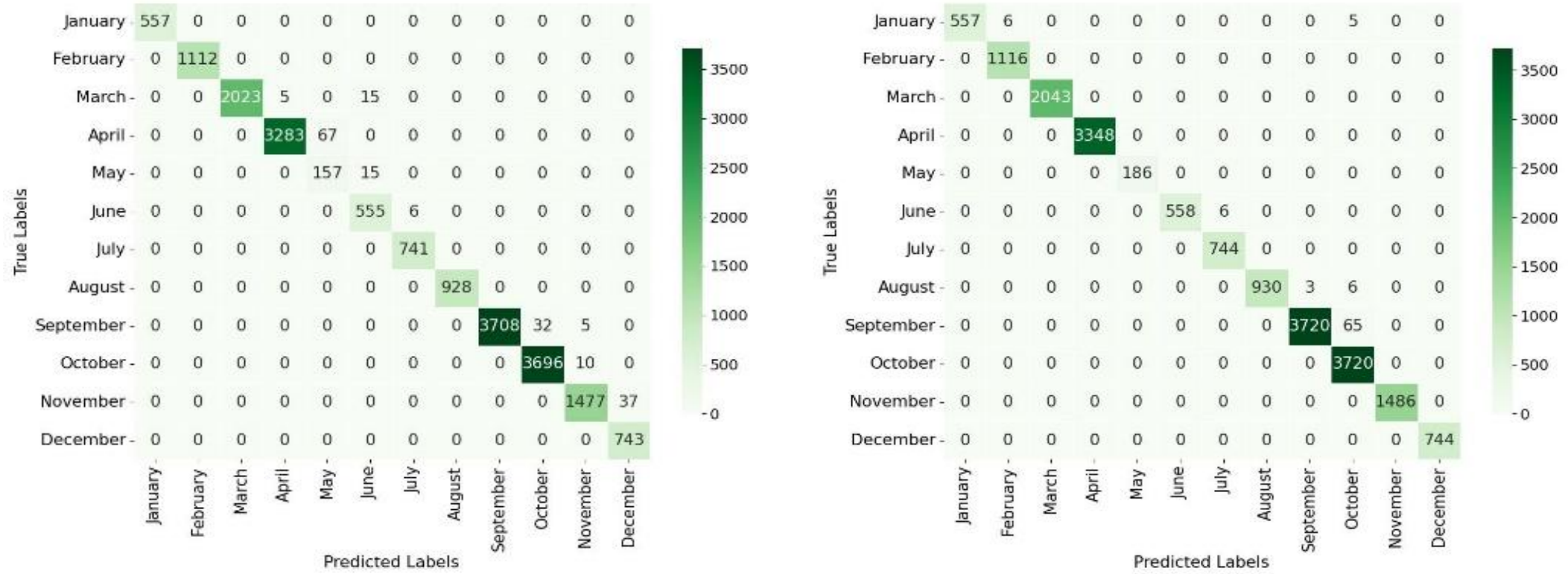


Fig. 4. Confusion Matrix for (a) KNN (b) RF.

MODEL VERIFICATION

$$\text{Zero Crossing Rate} = \frac{3720}{20340} \approx 18.29 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Model/Baseline	Accuracy	Classification Goal
Random Forest	98.95%	Classification across 12 months
KNN	97.68%	Classification across 12 months
Zero Rate (ZeroR) Baseline	18.29%	Always predicting the majority class (September/October)

Table 6. RF and KNN Performance Against Baseline.

MODEL VERIFICATION

Month	Support Count	RF F1-Score	KNN F1-Score	Insight
September (High Support)	3270	1.00	0.99	High performance on the majority class
May (Lowest Support)	186	1.00	0.87	RF generalization is superior for the severe minority class.
June (Low Support)	558	0.99	0.99	Both models perform well on this low-support class.

Table 7. High and Low support class F1-score Comparison.

CONCLUSION

- Our Random Forest–based model achieved 98.95% accuracy, effectively capturing the complex and climate-dependent patterns that determine optimal crop transplanting periods.
- Although highly accurate, the model currently predicts only the transplanting month, highlighting the need to incorporate yield, soil fertility, and satellite-based vegetation indices to build a more comprehensive decision-support system.
- Future advancements should integrate near-real-time weather data and hybrid deep-learning forecasts (e.g., CNN-LSTM-GRU) to deliver dynamic, real-time agro-advisories for climate-resilient agriculture.

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